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S2.3 The metallic model

Strength of the metallic bond

Learning outcomes

By the end of this section you should be able to:

- Predict the relative strength of the metallic bond by considering the charge and radius of the metal ion.
- Explain trends in melting points of s and p block metals.

Recall that a metallic bond is the electrostatic attraction between cations in a lattice and delocalised electrons, as illustrated in **Figure 1** below. By considering the particles that make up the metallic bond, can you hypothesise what factors would influence its strength? What factors might lead to a strong metallic bond? What factors might lead to a weak metallic bond? How would you expect the strength of a metallic bond to impact the melting point of a metal?

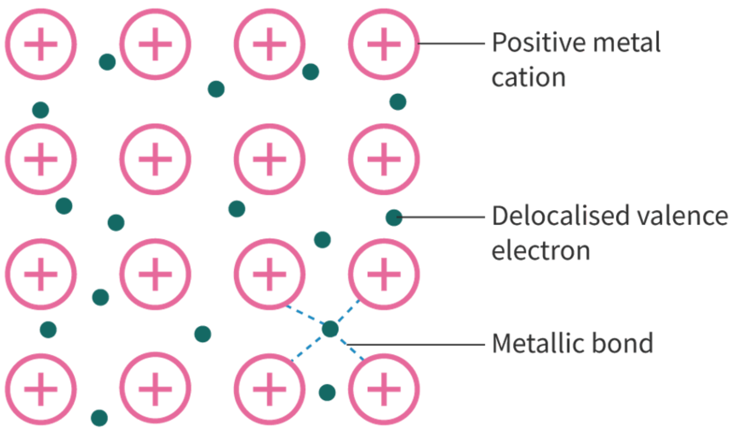

Figure 1. Metallic bond.

More information for figure 1

The diagram illustrates the structure of a metallic bond. It shows an orderly array of pink circles, each marked with a plus sign representing positive metal cations. Interspersed between these cations are small green circles representing delocalized valence electrons. Arrows and labels point out the metallic bond, which is the interaction between the cations and the electrons. The metallic bond is characterized by lines connecting the electrons to the surrounding cations, indicating the electrostatic attraction.

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Strength of the metallic bond

The strength of metallic bonds varies depending on the nature of the metal. The strength of the electrostatic force of attraction between delocalised electrons and cations within the metallic lattice is dependent on the same factors as those affecting ionic bonding, as in section S2.1.3a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/). Those factors are radius of the metal ion and ionic charge, which in combination determines the electron density. Let us consider the reasoning behind why these factors both influence metallic bond strength below.

Radius of metal ion

When you looked at the subatomic composition of the atom in section S1.2.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/) you may have noticed that the atom is mostly empty space. What do you think determines the radius of a metal cation? The protons in the nucleus could be very close or very far away from the valence shell electrons that delocalise through the metallic lattice. Do you think larger or smaller cations would have a stronger attractive force with the delocalised electrons in a metallic lattice?

The smaller the radius of the metal ion, the stronger the metallic bond. This is because of the shorter distance between the positive nucleus of the cation and the surrounding delocalised electrons.

Let us consider the strength of the metallic bond by considering the electron configuration and ionic radius for group 1 metals.

Table 1. Electron configuration and ionic radii for group 1 metals.

Element	Electron configuration	Charge of cation	Ionic radius (10^{-1}
Li	$[He]2s^1$	1+	76
Na	$[Ne]3s^1$	1+	102
K	$[Ar]4s^1$	1+	138
Rb	$[Kr]5s^1$	1+	152
Cs	$[Xe]6s^1$	1+	167

For group 1 metals, each atom loses one electron to the delocalised electrons in the lattice. Group 1 metal ions will have the same charge (1+) and the same number of delocalised electrons per metal atom. Metals further down the group will have a larger cationic radius due to the presence of an additional energy level.

As lithium metal ions have a smaller radius, the lithium nucleus will be closer to the delocalised electrons than the other group 1 metals and will result in a greater electrostatic attraction. This greater attraction means a stronger metallic bond, requiring more energy to break. Lithium, therefore, has the strongest metallic bond of the group 1 metals. This is illustrated in **Figure 2** below.

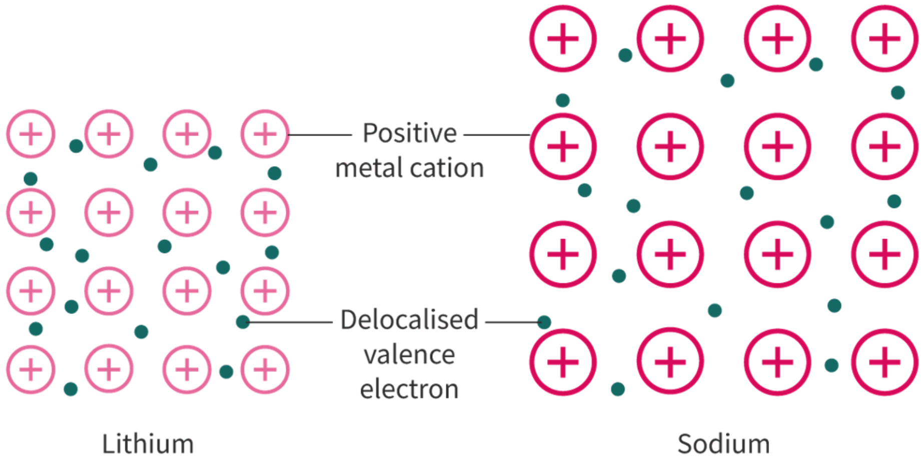

Figure 2. Comparison of the metallic bond of lithium and sodium.

More information for figure 2

The diagram compares the metallic bond structure of lithium and sodium. On the left, lithium is shown with a denser arrangement of positive metal cations and delocalized valence electrons. Positive metal cations are represented as circles with a positive sign inside, which are closely packed. Delocalized valence electrons are shown as small dots scattered between the cations. The labeling indicates that the distance between lithium's nucleus and its delocalized electrons is smaller, leading to a greater electrostatic attraction.

On the right, sodium is displayed with a similar structure but with a looser packing of positive metal cations and delocalized valence electrons, illustrating a lower electrostatic attraction compared to lithium. The comparison visually demonstrates why lithium forms a stronger metallic bond than sodium due to its smaller atomic radius, resulting in a greater attraction between the ions and electrons.

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Charge of the metal ion

Recall that the metallic bond is the electrostatic attraction between cations in a lattice and delocalised electrons. The charge of the cations is dependent on the number of electrons in the valence shell. How do you think the charge on the metal ion would impact the attractive forces with the delocalised electrons in the metallic lattice?

The higher the ionic charge, the stronger the metallic bond. This is because:

- greater charge on the metal ion

- greater number of delocalised valence electrons.

This greater charge difference results in a stronger electrostatic attraction and a stronger metallic bond.

Let us consider period 3 metals sodium and magnesium as an example.

Table 2. Electron configuration and charge for sodium and magnesium.

Element	Electron configuration	Charge of cation
Na	$[Ne]3s^1$	1+
Mg	$[Ne]3s^2$	2+

The ionic radius sizes for these two metal ions will be similar as they have the same number of occupied energy levels. Because magnesium has a 2+ charge, it will have a greater number of electrons making up the surrounding delocalised electrons than sodium. This greater charge difference increases the strength of the electrostatic attraction between metal ions and delocalised electrons. Magnesium, therefore, has a stronger metallic bond than sodium. This is illustrated in **Figure 3** below.

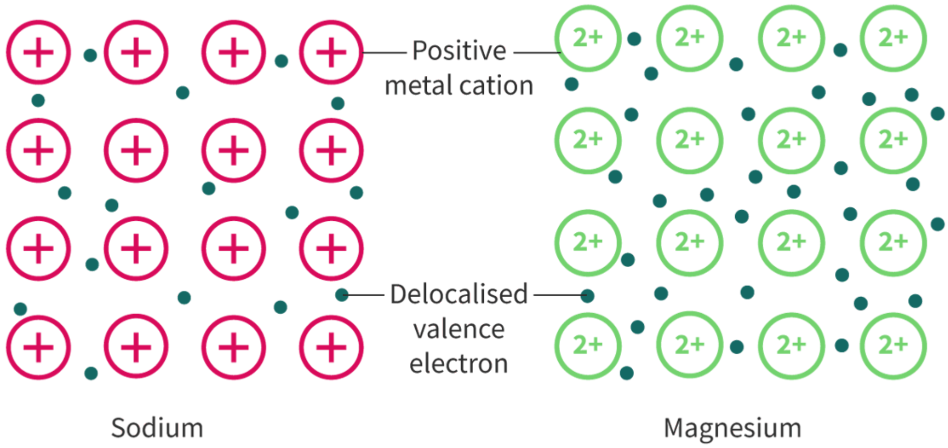

Figure 3. Comparison of the metallic bond of sodium and magnesium.

More information for figure 3

The image is a diagram comparing the metallic bonding structures of sodium and magnesium. On the left, sodium is depicted with '+' symbols inside circles representing positive metal cations, surrounded by small dots that represent delocalised valence electrons. On the right, the magnesium structure is similar but with '2+' inside the circles, indicating a stronger positive charge resulting in more delocalised electrons around the metal cations. This visual highlights the difference in the strength of the metallic bond, with magnesium showing a stronger attraction due to the increased charge and electron density.

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In summary, the metallic bond becomes stronger with increased electron density. Decreasing ionic radius and increasing charge on the metal ion increases the relative number of electrons in a region of space. The smaller the radius and the greater the charge of the metal ion, the stronger the metallic bond. The larger the radius and the lower the charge of the metal ion, the weaker the metallic bond. Try out **Interactive 1** below to see the effect of radius and charge of the metal ion on the strength of the metallic bond.

Adjust the radius or charge bar for Metal 2 and look at what happens to the metallic lattice image when you adjust the size, charge and the number of electrons.

Interactive 1. Effect of radius and charge of the metal ion on the strength of the metallic bond.

More information for interactive 1

An interactive simulation features two metals, Metal 1 and Metal 2. The cations in both metals initially have a +1 charge and the same number of delocalized electrons per metal atom within their metallic lattice.

Radius and Charge slider bars are at the bottom right of metal 2. The radius or the charge for Metal 2 can be manipulated to identify its influence on the metallic bond.

On moving the radius slider, the ionic radius of Metal 2 increases, and the displayed text reads Metal 1 has stronger metallic bonds than Metal 2. This explains how the increase in ionic radius affects the metallic bond.

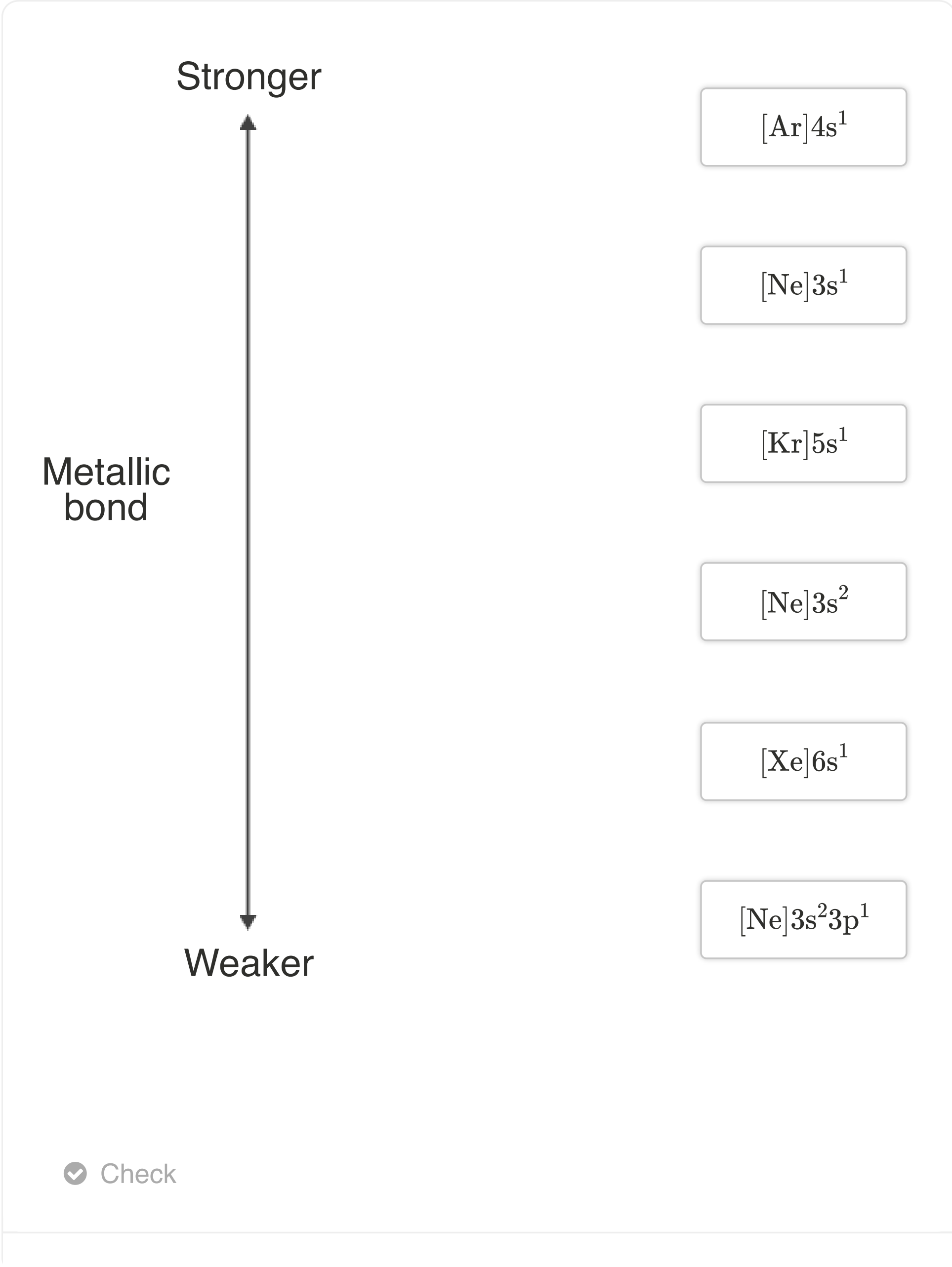
On moving the charge slider, the ionic charge of Metal 2 increases with the increase in the number of delocalized electrons, and the displayed text reads, Metal 1 has weaker metallic bonds than Metal 2. This explains how the increase in charge affects the metallic bond.

This interaction helps users understand the influence of metal ion radius and charge on metallic bond strength.

Making connections

The strength of the metallic bond is dependent on the same factors as the strength of the ionic bond, as explored in section S2.1.3a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/). The strength of both bonds depends on ionic radius and ionic charge.

Check your understanding of the strength of metallic bonds with **Interactive 2** by dragging and dropping the boxes of different electron configurations of metals and placing them in order of the strength of metallic bond from strongest to weakest.



Interactive 2. Understanding the Strength of Metallic Bonds.

Strength of metallic bond and melting points

The strength of metallic bonds results in high melting points for metals. From your experience, what state of matter are metals at room temperature? Do you know of any metal that exists in a different state at room temperature? Why might that be?

At room temperature, all metals, except mercury, are solids. For metals found in the s and p block regions of the periodic table, the melting point is directly related to the strength of the metallic bond. Note that the metals in the d block are not considered in this section due to the added complexity when considering the valence electrons in d orbitals. Melting points for all elements can be found in section 8 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/melting-points-and-boiling-points-of-the-elements-id-45110/) of the DP chemistry data booklet.

Figure 4 shows a periodic table with melting points for the s and p block metals highlighted. Looking at groups 1 and 2, what conclusion could you draw about the melting point of metals as you go down the group? Why do you think this is? Looking at period 3, what conclusion could you draw about the melting point of metals as you travel across a period? Why do you think this is?

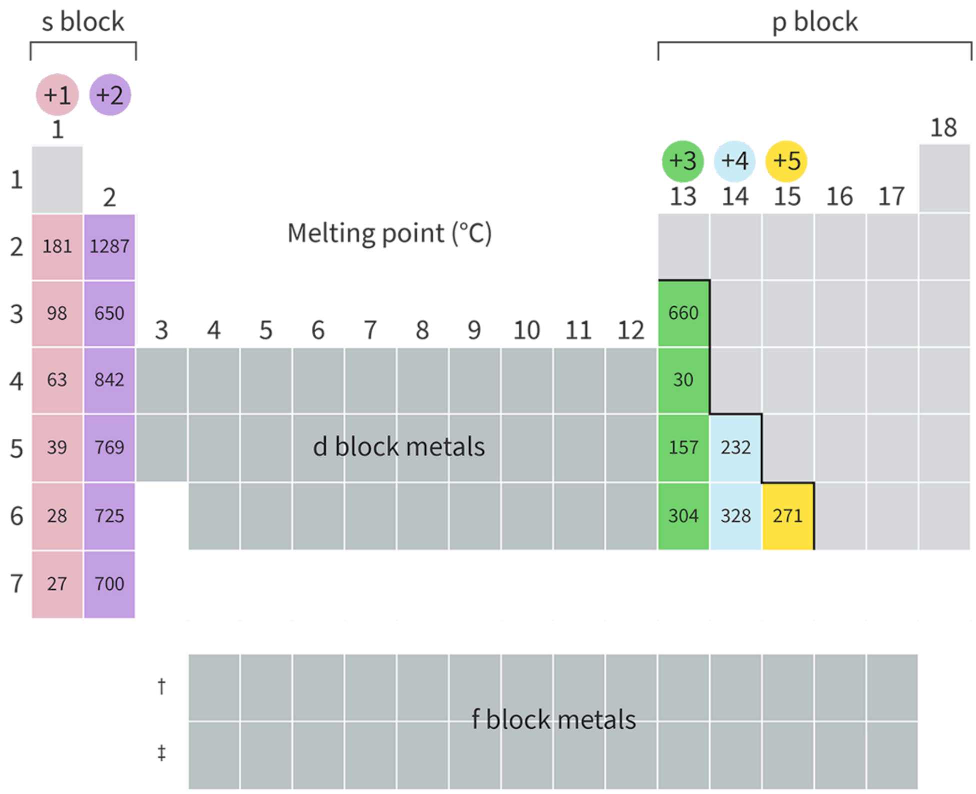


Figure 4. Melting points for the s and p block metals.

More information for figure 4

The image is a modified periodic table highlighting the melting points of s and p block metals. The table showcases groups 1 and 2, labeled with their respective changes in melting points as you move down the groups. For instance, Group 1 has melting points such as 181, 98, 63, 39, 28, and 27 moving downwards, indicating a general decrease. In Group 2, the melting points are 1287, 650, 842, 769, 725, and 700 moving downwards, showing variations. The s block metals are highlighted on the left side in shades of pink and purple. The p block metals on the right side include melting points such as 660, 30, 157, 304, 328, 232, and 271 in green, blue, and yellow blocks for typical groups +3, +4, and +5 respectively. The d and f blocks are shown in grey but not highlighted as they are not the primary focus. This visual representation indicates that melting points decrease downwards in groups and vary across periods due to differences in metallic bonding and electron density.

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As you can see, the melting point decreases as you go down a group of elemental metals. This is due to the decreased strength of the metallic bond as the radius of the metal ion increases. Generally, across a period, for elemental metals, the melting point increases. This is due to the increased strength of the metallic bond due to the increase in electron density as the charge on the metal ion becomes larger.

You might notice that the trend in melting point is much less distinct across a period than down a group. If you look at period 4, gallium is a significant outlier with a very low melting point: to the extent that objects made from gallium can melt when placed in the palm of your hand, or in warm water, as shown in **Video 1** below.

Video 1. Gallium spoon melting.

More information for video 1

The video presents a demonstration of a physical property of gallium metal. A transparent glass mug filled halfway with water is placed on a wooden table, alongside a polished, silver-coloured teaspoon made of gallium.
A hand enters the frame and lifts the spoon, initially holding it horizontally before rotating it to a vertical position. The hand then carefully lowers the spoon into the glass of water, immersing it. As the immersion progresses, the spoon is swirled around in the water. Over time, subtle changes become visible: small droplets begin to form along the spoon's surface. These droplets gradually increase in size before detaching and falling away, revealing that the spoon is undergoing a change. Rather than remaining solid, the metal appears to be melting. Eventually, the entire spoon melts completely, disappearing into the water.
The interactive video illustrates the low melting point of gallium as the metal transitions from a solid to a liquid when immersed in a cup with water. Gallium has a very low melting point, and when placed in water that is even slightly warm or on the palm of a hand, it begins to liquefy.
By showcasing the transformation of gallium in real time, the video highlights how physical properties such as melting point can vary significantly between different elements.

Theory of Knowledge

Mercury was consumed by many powerful historical figures, believing that it would grant immortality. Today, we have a greater understanding of this mysterious substance as a lethal neurotoxin, resulting in symptoms such as insomnia, emotional changes and disturbances in sensations. Does our current knowledge of this dangerous substance change our perspective of historical events?

Study skills

While there are trends for melting point in certain regions of the periodic table, there are not any trends for melting point across the periodic table. This is important to consider. The melting point trends discussed here are valid for s and p block metals, but do not apply to non-metals. It is important to consider the structure and bonding of elements to explain any trends.

Strength of metallic bond and hardness

The strength of metallic bonds results in a greater hardness of the metal. Hardness is a measure of the ability of a material to resist deformation. Think of the metals you have encountered at home: would you consider them hard or soft? What do you think that implies about the strength of their metallic bonds? Did you know that some metals are so soft they can be cut with a knife? **Interactive 3** shows two group 1 metals being sliced, sodium and potassium. Before watching them be cut, make a prediction. Which metal do you expect to be harder and have stronger metallic bonds? Which metal do you expect to be softer and have weaker metallic bonds? Then watch each metal being cut to see if you can identify the softer metal (weaker metallic bonds).

Interactive 3. Cutting Sodium and Potassium, a Measure of Hardness.

More information for interactive 3

The interactive contains two videos titled Sodium and Potassium. The video on the left for sodium is titled Sodium metal is soft and squishy. The video demonstrates the physical properties of sodium metal. The sodium metal chunks are stored in a glass container filled with oil. A chunk of sodium is removed from the oil and is pressed with fingers, revealing no signs of crumbling or breaking. The chunk is then cut using a knife. The video shows that sodium is soft enough to be easily cut with a knife, revealing its shiny interior. The cut piece is stored in the glass container with oil again. The video on the right for potassium is titled Potassium metal is like weird butter. The video demonstrates the physical properties of potassium metal. Potassium metal is taken out of a small tube and is then cut with a knife. The video shows that potassium is soft and can be easily cut like butter, revealing a shiny interior. The metal is pressed with fingers, revealing signs of crumbling and breaking. The cut piece of metal, when added to a container with water, reacts immediately emitting smoke and flame. The interactive demonstrates the metallic nature of sodium and potassium.

Activity

- **IB learner profile attribute**: Communicators
- **Approaches to learning**: Communication skills — Clearly communicating complex ideas in response to open-ended questions
- **Time required to complete activity:** 30—40 minutes
- **Activity type:** Individual activity

Spreadsheet plotting

In this activity you will be creating individual plots to show how the radius of the metal ion affects the melting point for group 1, group 2 and group 13 metals.

- Predict what you expect the shape of each graph to look like for each group. Create a sketch of what you think each graph will look like.
- Create individual plots showing the effect of the radius of the metal ion on the melting point for group 1, group 2 and group 13 metals using information found in section 8 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/melting-points-and-boiling-points-of-the-elements-id-45110/) of the DP chemistry data booklet.
- For each plot, consider the following questions:
    - Did this plot look similar to your sketch? Why or why not?
    - Are there any data points that did not align with your expectations? Try researching to see whether you can come up with any explanations.
    - How do the melting points of s and p block metals provide evidence for the strength of the metallic bond?